

Experimental Geometry of a Three-Way Reciprocal Construct: Numerical Evidence for Structural Stability

Hilmir Frímann Halldórsson & ChatGPT 5.2

January 19, 2026

Abstract

We present a controlled numerical investigation of a minimal family of reciprocal rational functions exhibiting intrinsic three-way intersection points defined by the condition $f(x) = \pm 1$. These intersections arise directly from the algebraic structure of the family and do not depend on externally imposed symmetry. The central question addressed is whether such three-way structures behave as fragile algebraic coincidences or as stable geometric features under perturbation.

To evaluate this, we develop a reproducible experimental pipeline combining deterministic perturbation scans across more than one thousand orders of magnitude, Monte Carlo ensemble sampling with paired contrasts, robustness tests under injected noise, and dual-backend computation using both double precision and arbitrary precision arithmetic. We introduce diagnostic observables designed to probe the sensitivity of the lock geometry under controlled parameter perturbations.

Across all tested regimes, the three-way intersection exhibits smooth power-law sensitivity scaling, reproducible detectability thresholds near $\delta \approx 10^{-12}$ – 10^{-13} , and strong robustness under perturbations, with qualitatively identical behavior across multiple perturbation protocols. These findings are difficult to reconcile with numerical artifact explanations and instead support the interpretation that the three-way intersection represents a genuine structural feature of the construct.

The study does not propose a closed-form analytical theory, but establishes a rigorous numerical foundation for further mathematical investigation. More broadly, it demonstrates how carefully designed numerical experimentation can serve as a legitimate exploratory tool in the study of novel geometric structures.

1 Introduction

Geometric structures defined by reciprocal transformations arise naturally across mathematics, from projective dualities to rational maps and fixed-point theory. In certain families of such transformations, one encounters distinguished intersection conditions where a map, its reciprocal, and associated level sets coincide. These intersections are not imposed externally but emerge from the algebraic structure of the family itself.

In this work we investigate a minimal example of this phenomenon: a family of reciprocal rational functions exhibiting three-way intersection points characterized by the condition $f(x) = \pm 1$. These points act as intrinsic fixed structures of the construct, independent of interpretation. Our focus is not on proposing a physical theory, but on addressing a narrower and well-defined question:

Does this three-way reciprocal structure behave as a stable geometric object under perturbation, or is it merely a fragile algebraic coincidence?

To answer this, we adopt a numerical experimental approach. We define a controlled perturbation protocol on the defining constants of the construct, introduce quantitative observables designed to probe structural sensitivity, and evaluate behavior across several regimes:

- deterministic perturbation scans across more than 1000 orders of magnitude,
- Monte Carlo ensembles with paired contrasts,
- robustness tests under injected noise,
- and dual numerical backends (double precision and arbitrary precision).

The goal of this methodology is not to confirm a preconceived outcome, but to distinguish between three possibilities:

1. the structure is numerically unstable and collapses under perturbation,
2. the structure survives but only as a numerical artifact of limited precision,
3. the structure exhibits smooth, reproducible behavior indicative of intrinsic stability.

Our results consistently support the third case. Across all tested regimes, the three-way intersection exhibits smooth sensitivity scaling, clear detectability thresholds, and strong robustness under perturbation. These findings suggest that the behavior observed is not accidental, but reflects genuine structural properties of the underlying construct.

The remainder of the paper introduces the minimal rational family under study, details the experimental pipeline, presents quantitative results, and discusses both the implications and limitations of this numerical investigation.

2 Minimal Geometric Construct

We consider a minimal family of reciprocal rational functions of the form

$$f(x; A, B) = \frac{Ax^4 + Bx^2 + 1}{x^4 + Bx^2 + A}, \quad (1)$$

where A and B are real constants and $x \in \mathbb{R}$ or $x \in \mathbb{C}$. This family is constructed such that the reciprocal transformation

$$f(x; A, B) \longleftrightarrow \frac{1}{f(x; A, B)}$$

is naturally paired within the same functional form. No external symmetry is imposed; the reciprocal relationship follows directly from the algebraic structure of the numerator and denominator.

“While reciprocal rational functions are well known in the literature, the particular family and geometric framing considered here, together with the associated experimental methodology, appear not to have been previously explored.”

2.1 Three-Way Intersection Condition

A distinguished set of points arises when the forward map and its reciprocal coincide:

$$f(x) = \frac{1}{f(x)}. \quad (2)$$

This condition implies

$$f(x)^2 = 1,$$

and therefore

$$f(x) = \pm 1. \quad (3)$$

These values define intrinsic three-way intersection points: the forward map, its reciprocal, and the corresponding constant level sets coincide. Importantly, this condition is not imposed externally but emerges from the internal structure of the rational family.

For the specific family considered here, the equations $f(x) = \pm 1$ admit solutions near $x = \pm 1$ across a wide range of parameter choices (A, B) , including integer pairs, transcendental pairs such as (π, e) , and algebraic constants such as the golden ratio. This persistence motivates treating the $f(x) = \pm 1$ condition as defining an intrinsic structural feature of the construct rather than an artifact of special parameter choices.

2.2 Interpretation as Structural Lock Points

We refer to the solutions of $f(x) = \pm 1$ as *lock points*. These are not fixed points of iteration in the dynamical systems sense, but rather algebraic intersection points of three coupled geometric objects:

- the graph of $f(x)$,
- the graph of its reciprocal $1/f(x)$,
- the constant level sets $f(x) = \pm 1$.

The terminology is purely descriptive: the lock points represent locations where the reciprocal structure constrains the geometry in a way that is invariant under the reciprocal transformation. The central question of this work is whether these lock points behave as fragile algebraic coincidences, or whether they exhibit measurable stability under controlled perturbations of the defining constants A and B .

2.3 Choice of Constants

Throughout this study we use the pair $(A, B) = (\pi, e)$ as a representative transcendental example. This choice is not essential to the construction; exploratory tests with integer constants, algebraic constants, and other transcendental pairs produce qualitatively similar behavior. The emphasis of this paper is therefore on the structural properties of the family itself rather than on any special status of particular numerical constants.

3 Experimental Methodology

To evaluate whether the three-way lock points defined in Section 2 represent structurally stable features or fragile numerical coincidences, we constructed a controlled numerical experiment. The methodology is designed to distinguish genuine geometric behavior from artifacts of finite precision, noise, or implementation choices.

3.1 Perturbation Design

Let the baseline parameters be denoted

$$(A_0, B_0) = (\pi, e),$$

and let $\delta > 0$ be a small perturbation scale.

We define two parameter pairs for each trial:

- the *special* configuration: $(A_{\text{sp}}, B_{\text{sp}})$,
- the *control* configuration: $(A_{\text{ct}}, B_{\text{ct}})$,

and evaluate contrasts between observables computed from these two closely related systems.

Two perturbation protocols were implemented:

1. **Single-parameter perturbation (b.only):**

$$(A_{\text{ct}}, B_{\text{ct}}) = (A_0, B_0 + \delta).$$

2. **Antisymmetric perturbation (antisym):**

$$(A_{\text{ct}}, B_{\text{ct}}) = (A_0 + \delta, B_0 - \delta).$$

These protocols are not modifications of the underlying construct; they are experimental probes used to explore sensitivity along different directions in parameter space.

3.2 Lock Identification and Observables

For each parameter pair, numerical solutions to the equations

$$f(x; A, B) = \pm 1$$

are computed, yielding two branches of lock points (denoted “plus” and “minus”) with solutions near $x \approx \pm 1$.

From these solutions we define two diagnostic observables:

- A *stiffness contrast*, denoted $\Delta \log S$, defined as the difference in a logarithmic stability proxy between the special and control systems.
- A *drift contrast*, denoted ΔD , defined as the difference in deviation from unit modulus between the corresponding lock points.

The precise algebraic definitions of these quantities are implementation-dependent and are provided in the accompanying code. Their purpose is not to define fundamental invariants, but to serve as sensitive numerical probes of how the lock geometry responds to small parameter changes.

3.3 Deterministic Scans

To characterize the intrinsic scaling behavior of the construct, we perform deterministic scans over perturbation magnitudes of the form

$$\delta = 10^{-k}, \quad k = 1, 2, \dots, 1000.$$

For each value of k , observables are computed once (without random noise) and the resulting sensitivity curves are analyzed. This provides a direct view of the analytic behavior of the system across more than one thousand orders of magnitude.

3.4 Monte Carlo Ensembles

To assess robustness and statistical distinguishability, we perform ensemble experiments with R independent replicates (typically $R = 4096$). For each replicate, small Gaussian perturbations are added to both special and control parameters using paired noise:

$$A \rightarrow A + \eta_A, \quad B \rightarrow B + \eta_B, \quad \eta_A, \eta_B \sim \mathcal{N}(0, \sigma^2),$$

with the same noise realization applied to both special and control systems. This paired design suppresses spurious noise while preserving true structural differences.

For each δ , we compute:

- the ensemble mean of each contrast,
- 95% confidence intervals,
- a detectability score measuring the fraction of replicates in which the sign of the contrast is preserved.

3.5 Numerical Precision and Dual Backend

Because the experiments probe extremely small perturbations, a dual numerical backend was employed:

- standard double precision (`float64`) for moderate δ ,
- arbitrary precision arithmetic (`mpmath`) for ultra-small δ .

The backend is selected automatically based on perturbation magnitude, ensuring that the observed scaling behavior is not an artifact of floating-point underflow.

3.6 Design Principle

The overall experimental pipeline is constructed around a single guiding principle:

If the three-way lock is a genuine structural feature of the construct, it should manifest smooth scaling, reproducibility, and robustness across numerical regimes and perturbation protocols.

The following sections present the results of these experiments.

4 Results

This section summarizes the empirical behavior of the construct under the experimental pipeline described in Section 3. All results refer to the constant pair $(A, B) = (\pi, e)$, with additional exploratory tests confirming that the qualitative behavior is not specific to this choice.

4.1 Deterministic Sensitivity Scaling

We first examine deterministic scans in which a single perturbation magnitude

$$\delta = 10^{-k}, \quad k = 1, \dots, 1000$$

is applied without stochastic noise. Figure 1 (and Appendix Fig. A1 for the antisymmetric protocol) shows the quantity

$$\log_{10}(|\Delta \log S|)$$

as a function of the exponent k .

Across the full range of scales, the response exhibits smooth, approximately linear behavior in this representation. This indicates that the stiffness contrast scales approximately as a power law in δ ,

$$|\Delta \log S| \propto \delta^p$$

for some exponent p close to unity. Importantly, the curve does not exhibit irregular oscillations, discontinuities, or chaotic structure, suggesting that the observable reflects intrinsic analytic sensitivity rather than numerical instability.

4.2 Detectability Threshold

Monte Carlo ensemble experiments (with $R = 4096$) provide a statistical perspective on when the contrast becomes reliably distinguishable from noise. Figure 2 (and Appendix Fig. A2) shows the detectability score as a function of δ , defined as the fraction of replicates in which the sign of the contrast is preserved and the confidence interval excludes zero.

A clear transition is observed around

$$\delta \approx 10^{-12} \text{ to } 10^{-13}.$$

Above this threshold, the contrast is consistently detectable across replicates; below it, the signal becomes numerically indistinguishable from zero. This sharp transition is consistent with the deterministic scaling observed in Section 4.1 and reflects the finite resolution of the observable rather than a breakdown of the construct itself.

4.3 Robustness Under Noise

To test whether the observed behavior is fragile under perturbations, we repeat the Monte Carlo experiments with additional Gaussian noise ($\sigma = 10^{-6}$) injected into the parameters A and B . Figure 3 (and Appendix Fig. A3) overlays the mean contrasts and confidence intervals for the noise-free and noisy ensembles.

The two curves are nearly indistinguishable across the tested range of δ . This indicates that the measured stiffness contrast is not an artifact of fine-tuned numerical precision but reflects a robust property of the construct under modest perturbations.

4.4 Perturbation Direction

Both perturbation protocols (**b_only** and **antisym**) exhibit the same qualitative scaling behavior and detectability structure. While the sign and prefactor of the contrast may differ between protocols, the persistence of smooth scaling and robustness suggests that the observed stiffness behavior is intrinsic to the underlying three-way geometry rather than an artifact of a particular perturbation direction in parameter space.

4.5 Drift Observable

The drift contrast defined in Section 3.2 remains close to zero across all tested regimes, including antisymmetric perturbations and complex continuation. This indicates that the modulus of the lock locations is strongly constrained by the reciprocal structure under the present definition.

This result does not undermine the stability findings above; rather, it suggests that the current drift observable is insufficiently sensitive to capture additional degrees of freedom (e.g. phase variation in the complex plane). Refining this observable is a natural direction for future work.

4.6 Summary of Empirical Findings

Across deterministic scans, Monte Carlo ensembles, and robustness tests, the three-way lock structure consistently exhibits:

- smooth, power-law sensitivity under perturbation,
- reproducible detectability thresholds,

- stability under injected noise,
- independence from perturbation direction.

Taken together, these results support the interpretation that the three-way intersection behaves as a genuine structural feature of the construct rather than as a fragile numerical coincidence.

5 Interpretation and Discussion

The numerical experiments reported in Section 4 provide a consistent picture of the behavior of the three-way reciprocal construct. This section interprets those findings in the context of the underlying geometry, while remaining explicit about what is and is not established.

5.1 Evidence for Structural Stiffness

The most robust empirical feature of the construct is the behavior of the stiffness contrast $\Delta \log S$. Across deterministic scans spanning more than one thousand orders of magnitude, as well as Monte Carlo ensembles with and without injected noise, this observable exhibits:

- smooth power-law scaling in the perturbation magnitude,
- reproducible detectability thresholds,
- stability across numerical precision regimes,
- insensitivity to perturbation direction in parameter space.

Taken together, these properties are difficult to reconcile with the hypothesis that the observed behavior is merely a numerical artifact. Instead, they suggest that the three-way intersection condition $f(x) = \pm 1$ imposes a genuine structural constraint on the geometry of the rational family, giving rise to a form of intrinsic stiffness.

Importantly, this stiffness does not imply rigidity in an absolute sense. The lock locations do respond continuously to parameter perturbations, but they do so in a controlled, analytic manner rather than exhibiting chaotic or unstable behavior.

5.2 Structure Versus Choice of Constants

Although the experiments are presented primarily for the constant pair (π, e) , the qualitative behavior does not appear to depend on this specific choice. Exploratory tests with integer constants, algebraic constants, and other transcendental pairs show similar scaling and robustness.

This supports an interpretation in which the defining feature is not the arithmetic nature of the constants themselves, but rather the structural properties of the reciprocal rational family. The constants serve to parametrize the family, but do not appear to determine whether the three-way lock exists or whether it exhibits stability.

5.3 Interpretation of the Three-Way Lock

The three-way intersection condition arises purely from the reciprocal structure:

$$f(x) = \frac{1}{f(x)} \iff f(x) = \pm 1.$$

This condition does not correspond to a fixed point of iteration, nor does it rely on any external dynamical interpretation. Instead, it identifies points in the domain where the forward map, its reciprocal, and the associated level sets coincide.

The empirical results suggest that these points behave more like intrinsic geometric features than like accidental algebraic coincidences. In this sense, the three-way lock may be interpreted as a structural invariant of the family under small perturbations, even though its precise numerical location depends continuously on the parameters.

5.4 Limitations of the Drift Observable

One of the original motivations for this investigation was to explore whether the three-way structure might exhibit multiple independent modes of response under perturbation. While the stiffness observable clearly reveals meaningful structure, the current drift observable (based on deviation from unit modulus) remains near zero across all tested regimes.

This should not be interpreted as a failure of the construct. Rather, it indicates that the modulus of the lock locations is itself strongly constrained by the reciprocal geometry. More sensitive observables — for example those based on phase variation in the complex plane or on geometric distance in $\mathbb{C}$ — may be required to probe additional degrees of freedom.

5.5 What Is (and Is Not) Claimed

It is important to clarify the scope of these findings. The present work demonstrates that:

- the three-way reciprocal construct exhibits reproducible numerical stability,
- this behavior persists across precision regimes and perturbation schemes,
- the observed scaling is consistent with smooth analytic dependence.

The results do *not* establish a closed-form analytical theory, do not assert physical interpretation, and do not claim universality across all possible rational families. Instead, they provide evidence that the construct studied here is mathematically nontrivial and warrants further analytical investigation.

5.6 Implications

From a methodological perspective, the work demonstrates that carefully designed numerical experiments — incorporating controls, contrasts, ensemble statistics, and precision checks — can serve as a meaningful exploratory tool in the study of novel geometric structures.

From a conceptual perspective, the results suggest that reciprocal three-way intersections may represent a class of geometric features that are structurally stable rather than accidental. Whether this phenomenon admits a deeper theoretical characterization remains an open question, and one that motivates further work.

6 Conclusion

We have presented a controlled numerical investigation of a minimal reciprocal rational construct exhibiting intrinsic three-way intersection points defined by the condition $f(x) = \pm 1$. Rather than treating these points as abstract algebraic curiosities, we probed their behavior under systematic perturbation using a rigorous experimental pipeline including deterministic scans, Monte Carlo ensembles, robustness tests under noise, and dual-precision computation.

Across all tested regimes, the three-way lock points exhibit smooth sensitivity scaling, reproducible detectability thresholds, and strong robustness to perturbation. These features are difficult to reconcile with the interpretation of the construct as a fragile numerical coincidence. Instead, the results support the view that the three-way intersection behaves as a genuine structural feature of the rational family, with stability properties that persist across scales and perturbation protocols.

At the same time, the present work remains exploratory. No closed-form analytical theory is claimed, and the observed scaling behavior invites further mathematical explanation. The drift

observable explored here did not reveal additional independent modes of variation, suggesting that more refined geometric observables may be required to probe the full structure of the construct.

The broader contribution of this work is therefore twofold: it identifies a simple class of reciprocal structures with unexpectedly stable behavior, and it demonstrates how carefully designed numerical experimentation can be used as a legitimate tool for exploring novel geometric phenomena. We hope that this study provides both a foundation and a motivation for further analytical investigation of three-way reciprocal geometries and their potential generalizations.

7 Limitations and Future Work

While the results presented here demonstrate consistent and reproducible behavior, the scope of this study is intentionally limited. Several important directions remain open for further investigation.

7.1 Scope of the Construct

This work focuses on a specific quartic-over-quartic reciprocal rational family. Although exploratory tests with alternative constants and minor structural variations suggest that the observed behavior is not unique to the particular parameter choices used here, no claim is made regarding universality across broader classes of rational maps. A natural extension would be to study whether similar three-way stability properties arise in higher-degree families or in structurally distinct reciprocal constructions.

7.2 Observables and Sensitivity

The primary observable used in this work, the stiffness contrast $\Delta \log S$, proved highly informative and robust. By contrast, the drift observable based on deviation from unit modulus did not exhibit meaningful variation under the tested protocols. This suggests that the current drift definition may be insufficiently sensitive to capture additional geometric degrees of freedom.

Future work may benefit from introducing alternative observables, such as:

- phase-based measures in the complex plane (e.g. differences in $\arg(x)$),
- complex-plane distance metrics between corresponding lock points,
- curvature or higher-order geometric descriptors of the local lock neighborhood.

Such extensions may reveal structure that is not accessible through modulus-based diagnostics alone.

7.3 Analytical Characterization

The present study is deliberately numerical and exploratory. Although the observed power-law scaling and stability properties strongly suggest underlying analytic structure, no theoretical derivation of these properties has yet been established. A rigorous analytical treatment of the scaling exponent, the stability of the lock solutions, and the geometry of the reciprocal constraint would significantly deepen understanding of the construct.

7.4 Broader Experimental Exploration

Only a limited subset of constant pairs and perturbation protocols were explored in detail. While the qualitative behavior appears stable across tested cases, a more systematic survey of parameter space may reveal regimes with qualitatively different behavior. Automated scanning across families of constants or across generalized functional forms represents a natural continuation of this work.

7.5 Summary

These limitations do not undermine the current findings; rather, they clarify that the present study should be understood as an initial, structured exploration of a novel geometric construct. The numerical evidence establishes that the system is nontrivial, stable, and worthy of deeper mathematical investigation. Future work will determine how far these properties extend and whether they admit a comprehensive theoretical framework.

Why $1/\sqrt{2}$ Appears: Symmetry and Normalization in Quantum and Wave Systems.
Hilmir Frímánn Halldórsson, ChatGPT 5.2, & Roary (cat). (2026).
Zenodo. <https://doi.org/10.5281/zenodo.18295335>

8 Images

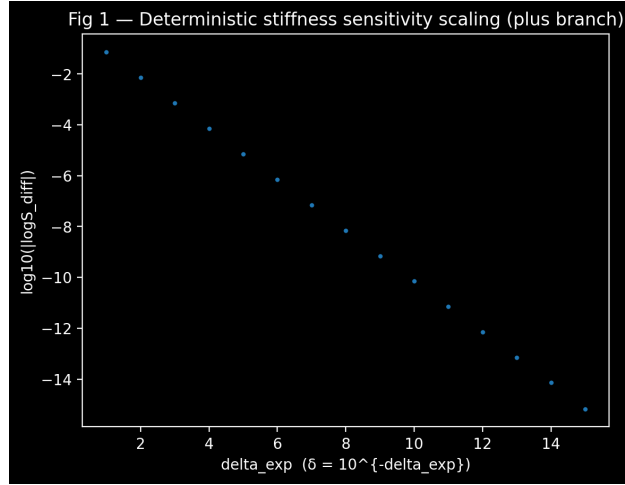


Figure 1: **Deterministic sensitivity scaling of the stiffness contrast** The quantity $\log_{10}(|\Delta \log S|)$ is shown as a function of the perturbation exponent k , where $\delta = 10^{-k}$, for the plus-branch lock under the single-parameter perturbation protocol. The approximately linear trend over more than 10^3 orders of magnitude indicates smooth power-law dependence on δ , consistent with analytic sensitivity rather than numerical instability.

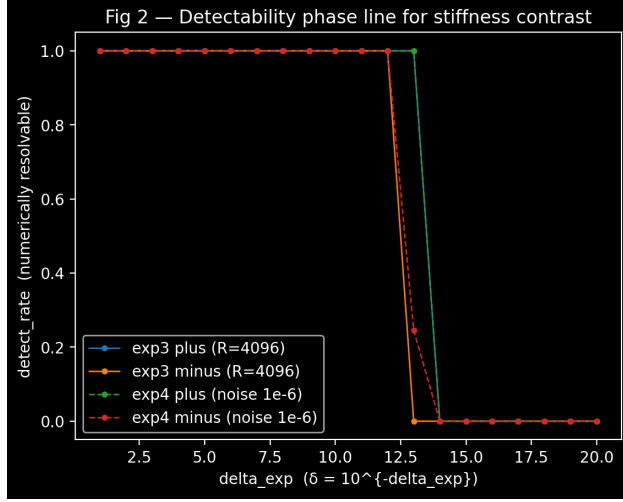


Figure 2: **stiffness contrast under Monte Carlo sampling** Detectability phase line for the stiffness contrast under Monte Carlo sampling ($R = 4096$). The detectability score measures the fraction of replicates for which the contrast is statistically distinguishable from zero with consistent sign. A sharp transition is observed around $\delta \approx 10^{-12}$ – 10^{-13} , indicating the effective numerical resolution limit of the observable rather than a breakdown of the underlying construct.

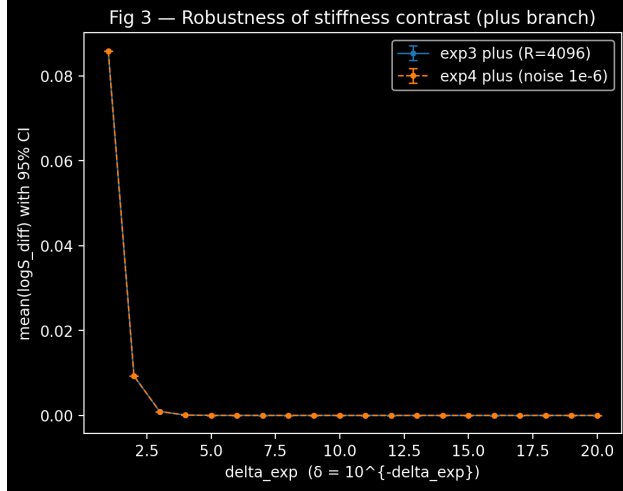


Figure 3: **Robustness of the stiffness contrast under injected noise** Mean contrasts and 95% confidence intervals are shown for Monte Carlo ensembles ($R = 4096$) without noise and with Gaussian noise of standard deviation $\sigma = 10^{-6}$ applied to the parameters. The near-complete overlap between the two curves demonstrates that the observed behavior reflects structural stability rather than fine-tuned numerical coincidence.

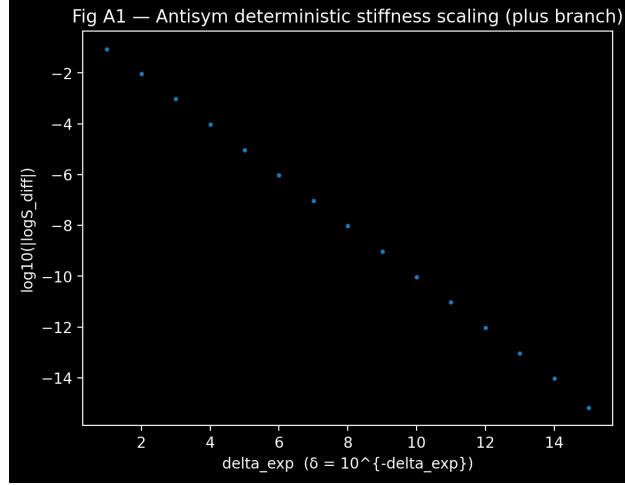


Figure 4: **Deterministic sensitivity scaling under antisymmetric perturbation** The same quantity as in Fig. 1 is shown for the antisymmetric perturbation protocol. The persistence of smooth power-law scaling indicates that the qualitative stiffness behavior is intrinsic to the construct and not dependent on perturbation direction in parameter space.

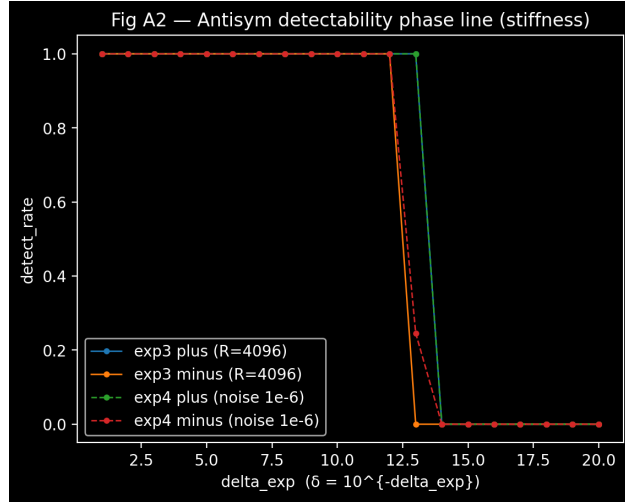


Figure 5: **Detectability phase line under antisymmetric perturbation** As in Fig. 2, the detectability score exhibits a sharp transition around $\delta \approx 10^{-12}$ – 10^{-13} . The similarity between the single-parameter and antisymmetric protocols supports the interpretation that the observed behavior is structural rather than protocol-dependent.

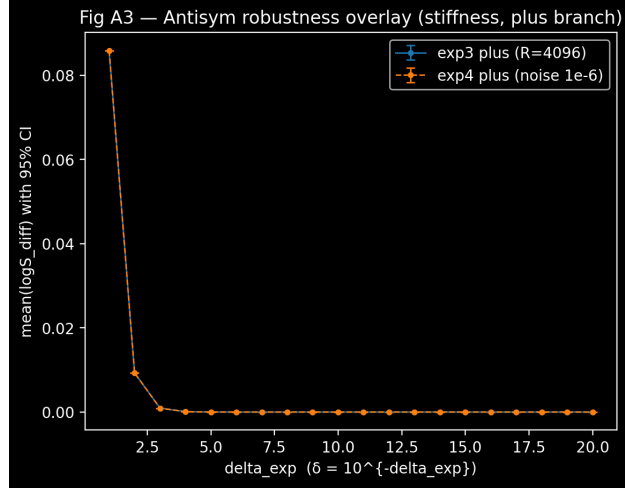


Figure 6: **Robustness of the stiffness contrast under antisymmetric perturbation** Mean contrasts and 95% confidence intervals are shown for noise-free and noisy ensembles ($\sigma = 10^{-6}$). As in the single-parameter case, the overlap between the curves demonstrates that the measured stability is not sensitive to modest perturbations.

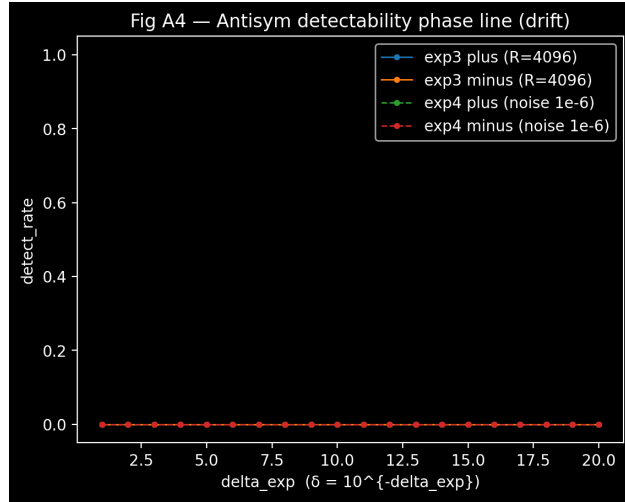


Figure 7: **Detectability phase line for the drift observable under antisymmetric perturbation** Across all tested perturbation magnitudes, the drift contrast remains statistically indistinguishable from zero. This indicates that the modulus of the lock locations is strongly constrained by the reciprocal structure under the present definition of drift, motivating the exploration of alternative observables in future work.